

ABHINAY PATTHI

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SUMMARY

Lead Software Engineer with **10+ years of experience** in **high-scale distributed systems** and **event-driven architectures**. Architects cloud-native solutions handling billions of records daily at sub-second latency. Integrates **Generative AI** and autonomous agents into operational workflows to eliminate engineering toil and drive developer productivity at enterprise scale.

TECHNICAL SKILLS

Languages: Python, Java

AI: GenAI, LLMs, Prompt Engineering, MCP

Cloud Platforms: AWS (EC2, S3, Lambda, RDS, SNS, SQS, DMS, API Gateway), Azure, GCP

CI/CD & DevOps: Jenkins, GitHub Actions, GitLab CI/CD

Message Queues & Streaming: Kafka, RabbitMQ

Architecture & Design: Distributed Systems, Event-Driven Architecture, System Design, Scalable Backend Architecture

Tools & Platforms: Linux, Unix, Postman, Jira, Confluence, Bruno

Frameworks: Spring, Spring Boot, FastAPI, Django

Databases: PostgreSQL, Redis, MongoDB, Elastic Search

Containerization & Orchestration: Docker, Kubernetes

Monitoring & Observability: Prometheus, Grafana, CloudWatch, Datadog

Version Control & Collaboration: Git, GitHub, GitLab, Bitbucket

Security & Authentication: JWT, OAuth 2.0, API Security, RBAC

WORK EXPERIENCE

Lead Software Engineer

Jan 2025 – Present

Financial Industry Regulatory Authority (FINRA)

New York, NY

- Built an **autonomous AI Agent** leveraging **GenAI** to automate Lambda runtime upgrades across the enterprise, cutting manual engineering effort by **80%** and enforcing continuous compliance with the latest AWS security patches.
- Architected a **serverless REST API** on AWS to handle high-concurrency data dissemination, sustaining **sub-second latency** under peak loads of **1800+ TPS** with zero degradation.
- Designed a fault-tolerant **data ingestion pipeline** using **Apache Kafka** and **Aurora PostgreSQL**, reliably processing **1.2 billion records daily** with high write throughput to meet enterprise-scale SLA requirements.
- Established observability infrastructure using **CloudWatch** and **Datadog** for distributed tracing, reducing **MTTD** for production incidents and improving system reliability across microservices.

Software Engineer

Feb 2019 – Dec 2024

Financial Industry Regulatory Authority (FINRA)

Rockville, MD

- Engineered an asynchronous **serverless RESTful API** using **S3 pre-signed URLs** for high-performance bulk data retrieval, minimizing server load and bandwidth costs.
- Implemented **Redis**-based caching and rate limiting across microservices, significantly reducing database load and improving API response times for high-frequency data access patterns.
- Built a standardized **Lambda Authorizer** for AWS API Gateway adopted enterprise-wide, reducing cross-team development cycles by **15%** with a plug-and-play authentication solution.
- Led migration of internal APIs to **AWS Fargate**, optimizing container orchestration and reducing infrastructure code by **62%**.

Software Developer Intern

Oct 2017 – Dec 2018

Senecio Corporation

Bowling Green, OH

- Built a high-performance **geospatial REST API** using **Redis Geohashing**, enabling real-time location-based data retrieval with **sub-500ms latency**.
- Engineered an asynchronous backend service for event creation, optimizing **multi-part upload** logic for high-volume media ingestion.
- Improved system responsiveness by offloading media processing to **background workers**, ensuring seamless UX during large file uploads.

Programmer Analyst

Jul 2014 – Dec 2016

Cognizant Technology Solutions

Pune, India

- Developed enterprise-grade web applications with **Spring Boot**, implementing architectural patterns for maintainability and scalability.
- Collaborated in a globally distributed **Agile** team to deliver modern web architectures with seamless cross-functional integration.

EDUCATION

Bowling Green State University

Master of Science in Computer Science

Government College of Engineering

Bachelor of Science in Computer Science

Bowling Green, OH, USA

Dec 2016 – Dec 2018

Amravati, Maharashtra, India

Jun 2010 – Apr 2014

CERTIFICATIONS

AWS Certified Developer – Associate ([View Certificate](#))